

Muhammad Tahir Zia

PASSCO Society, Lahore, Pakistan | itstahir256@gmail.com | [+92-304-5744237](tel:+92-304-5744237)

[Linkedin.com/in/Muhammad-Tahir-Zia](https://www.linkedin.com/in/Muhammad-Tahir-Zia) | [GitHub.com/TahirZia-1](https://github.com/TahirZia-1) | [Portfolio](#)

Summary

Aspiring Digital IC Design & Verification Engineer with hands-on experience in SystemVerilog RTL design, RISC-V microarchitecture, and verification. Proficient in RTL design and simulation. Passionate about FPGA prototyping for complex digital systems.

Education & Training

Digital IC Design & Verification (INSPIRE) Bootcamp, Pakistan Software Export Board (PSEB) at GIKI May 2026 – Present

- **Focus:** Intensive specialized training in RTL design, advanced processor microarchitecture, and functional verification methodologies.

BS in Computer Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan Sept 2021 – June 2025

- **Coursework:** Fundamentals of Logic Design, Computer Organization and Assembly Language, Microprocessor Interfacing, Computer Communication and Networking, Signals & Systems, Digital Systems Design.

Technical Skills

Languages & Standards: SystemVerilog, Verilog, C/C++, RISC-V Assembly.

Verification & Design Concepts: SystemVerilog Testbenches, FSM Design, Pipelining, Clock Domain Crossing (CDC).

EDA Tools & Platforms: AMD Vivado, Icarus Verilog, EDA Playground, Nexys A7 FPGA, Cadence Xcelium.

Projects

32-bit RISC-V Pipelined Processor Core – *SystemVerilog, Vivado* ([GitHub Repository](#))

- Architected an RV32I 5-stage pipelined processor, evolving from a foundational single-cycle datapath.
- Engineered a custom Hazard Unit to dynamically resolve data and control hazards via EX/MEM & MEM/WB forwarding, load-use stalling, and branch flushing.
- Working on a branch predictor by utilizing 8-bit Global History Register and 256-entry Pattern History Table.

FPGA Morse Code Encoder – *Verilog, Vivado, Nexys A7* ([GitHub Repository](#))

- Converts digits into its Morse code using an FSM based approach, displayed using LED lights. 1 second for dot and 3 seconds for Dash. Can be toggled into different modes for storing numbers in FIFO register as well.

Digital Clock – *Verilog, Vivado, Nexys A7* ([GitHub Repository](#))

- Verilog implementation of a 24-hour digital clock. The design displays hours, minutes, and seconds on a 7-segment display that can be easily integrated into various FPGA development boards like Nexys A7.

FPGA Lab Design & Verification Work ([GitHub Repository](#))

- Designed and verified a **4-bit array multiplier** and **6-bit CSA multiplier** in Verilog.
- Implemented Parallel Prefix Adders including **Kogge-Stone**, **Sklansky**, and **Brent-Kung** adders.
- Designed FSM-driven signal generators supporting **Short Blink** and **Long Blink** modes on the Nexys A7 FPGA.

Final Year Project

BlockConnect (Social Media Application using Blockchain) May 2024 – Jun 2025

- Developed a decentralized social networking layer by using Ethereum blockchain focused on data sovereignty and user privacy. Leveraged the architecture to eliminate the third-party data commoditization.